

Big Data and Society

**Class041 Affiliation
Networks**

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Affiliation networks (also called **two-mode networks** or **bipartite networks**) are a type of social network where **two distinct types of nodes** are connected—such as *people* and *organizations*, *users* and *hashtags*, *investors* and *companies*, or *consumers* and *products*. 隸屬網路（也稱為二模網路或二分網路）是一種社交網路，其中連接的是兩類不同的節點——例如人與組織、使用者與話題標籤（hashtags）、投資者與公司，或消費者與產品。

Unlike traditional (one-mode) networks that link similar entities (e.g., person-to-person), affiliation networks capture **membership**, **participation**, or **co-occurrence relationships**. 與傳統的（單模）網路（如人與人之間的連接）不同，隸屬網路刻畫的是成員關係、參與關係或共現關係。

Structure of an Affiliation Network

- **Mode A:** Actors (e.g., users, investors, customers) 行動者（例如用戶、投資者、顧客）
- **Mode B:** Affiliations (e.g., groups, brands, stocks, events) 隸屬物件（例如群體、品牌、股票、活動）
- **Edges:** Exist only between Mode A and Mode B (never within the same mode) 僅存在於模A與模B之間（同一模內部不存在連接）

Application in Marketing

Building on **bipartite (affiliation) networks**, we can construct powerful tools for retail, e-commerce, and marketing—especially in **cross-selling** and **product recommendation**. 基於二分（附屬）網路，我們可以構建強大的工具應用於零售、電子商務和行銷領域——特別是在 **交叉銷售** 和 **產品推薦** 方面。

From Bipartite Network to Product Affinity Network 從二分網絡--> 產品親和網絡

Start with a **Bipartite Customer-Product Network**

- **Mode A:** Customers (e.g., $C_1, C_2, \dots, C_n$) **顧客**（消費者、企業客戶、APP使用者）是一類節點 and **Mode B:** Products (e.g., $P_1, P_2, \dots, P_m$) **產品**（如手機、咖啡、保險套餐、短視頻內容）是另一類節點
- **Edge:** Exists if a customer purchased (or viewed/rated) a product. 當顧客購買、流覽、收藏或互動某產品時，即建立連接。

- Once you have established an edge between Product A and Product B, what does that relationship actually mean business-wise?
 - Complements (The Most Common Case) : Bundle them, place them near each other, or recommend B when A is in the cart

Example: $C_1 \rightarrow P_1, P_3$; $C_2 \rightarrow P_1, P_2, P_3$; $C_3 \rightarrow P_2$

- 例如：
 - User A purchased a coffee machine and coffee beans → creating links: A – Coffee Machine, A – Coffee Beans 用戶A 購買了 咖啡機 和 咖啡豆 → A – 咖啡機，A – 咖啡豆
 - User B browsed an electric toothbrush and a water flosser → implying a demand association within the oral care category. 用戶B 流覽了 電動牙刷 和 沖牙器 → 暗示口腔護理品類存在需求關聯。

Project to a **Product Co-Purchase Graph** (One-Mode Projection). We now create a **Product Affinity Network** — a weighted graph 加權的產品親和網路 (Product Affinity Network) 將原始的顧客-產品二分網路投影為產品-產品網路 where:

- **Nodes:** Products only. 節點：所有產品; **Edges:** Exist between two products if **at least one customer bought both**. 若至少一位顧客同時接觸（購買/流覽）兩個產品，則連邊
- **Edge weight:** Number of customers who co-purchased the pair. 邊權重：共現頻次 或 相似度得分

From above: P_1-P_3 : 2 customers (C_1, C_2); P_1-P_2 : 1 customer (C_2); P_2-P_3 : 1 customer (C_2). This graph reveals **product affinities**—natural groupings based on real behavior. 這就是電商平臺“經常一起買”、“看了又看”功能的演算法基礎

- Product co-purchase : 成對購買

Applications: Cross-Selling 交叉銷售 (Cross-Selling) & Recommendations

The **Product Affinity Network directly** enables: 產品親和力網絡可以

- **"Frequently bought together"** suggestions (Amazon-style). 成對購買的頻率
- **Cross-selling:** Recommend Product B when a user views/buys Product A. 交叉銷售：基於 product 親和網路，在使用者購買A時推薦 高親和度 的B（如買手機推薦耳機）；
- **Category discovery:** Identify **substitute** or **complementary** products.

向上銷售 (Up-Selling)

- **向上銷售：**Identify "upgrade paths"—for example, users who purchase an entry-level camera often buy professional lenses within three months, allowing for proactive promotional triggers. 識別“升級路徑”——例如，購買入門款相機的用戶，常在3個月內購買專業鏡頭，可提前觸發促銷。

But simple co-purchase counts can be misleading (e.g., popular items co-occur by chance). That's where **similarity metrics** help. 簡單只看共同購買是不夠的，必須要從 相似性 著手

Key Similarity Metrics for Product Affinity 產品親和下的相似度指标

1. Common Neighbors (CN) (共同鄰居)

- **Count** of shared customers between two products. 同時購買 產品A和B的顧客數量
- **Formula:** $CN(P_i, P_j) = | N(P_i) \cap N(P_j) |$, where $N(P)$ = set of customers who bought product P.
- Simple and intuitive. ❌ Biased toward popular products. 快速識別強關聯組合 (如尿布+啤酒)

2. Jaccard Similarity

- What fraction of *unique buyers* purchased both products? 通過計算 兩個商品 被同一顧客共同購買的頻率，衡量它們的關聯強度
- **Normalizes** common neighbors by total unique buyers of either product. 共同顧客數 / (購買A 或 B的總顧客數)
- **Formula:**
$$\text{Jaccard}(P_i, P_j) = \frac{|N(P_i) \cap N(P_j)|}{|N(P_i) \cup N(P_j)|}$$
- Reduces popularity bias.  Values between 0 and 1.
 -  Still sensitive when sets are small. **抑制** 熱門產品干擾，發現 **小眾** 但高契合的搭配（如高端鏡頭+濾鏡）

Example: How similar are Coffee (A) and Sugar (B) ?

Buyers of Coffee (A) = {Alice, Bob, Carol} and Buyers of Sugar (B) = {Alice, Bob, Dave}

--> Intersection $A \cap B = \{\text{Alice, Bob}\} \rightarrow 2$ customers bought BOTH, Union $A \cup B = \{\text{Alice, Bob, Carol, Dave}\}$

$\rightarrow 4$ customers bought EITHER, finally, $\text{Jaccard}(A, B) = |A \cap B| / |A \cup B| = 2 / 4 = 0.5$ (or 50%)

Jaccard Similarity filters out the 'noise' generated by bestsellers, focusing instead on combinations with a genuinely high co-occurrence **ratio**. Jaccard相似度：排除熱門商品的“噪音”，聚焦真正共現比例高的組合. A high Jaccard value indicates a specific, **high-frequency co-purchase pattern**, revealing true product affinity rather than general popularity Jaccard值高，說明存在 **特定商品們的高頻共現**

3. Adamic-Adar (AA) Index

- Weights shared neighbors by **inverse log frequency**: rare co-buyers contribute more. 對每個共同顧客，按其 總購買品類數加權（越專注的顧客權重越高）
- **Formula**: $AA(P_i, P_j) = \sum_{c \in |N(P_i) \cap N(P_j)|} \frac{1}{\log |N(c)|}$, where $|N(c)|$ = number of products customer c bought
- Rewards connections through **niche or focused buyers** (more meaningful signal). 
Effective at filtering noise from "heavy buyers" who purchase everything. 優先採納“**重度**興趣使用者”的行為信號，提升推薦精準度

Adamic-Adar指數: A greater weight on the co-purchase behavior of "**low-activity but highly focused**" customers. 更重視“低活躍但專注”的顧客的共現行為。

If two products are bought together by a customer who only buys 2 items, that's more significant than if they're bought by a customer who buys 100 items. 如果 兩位產品 是被一位僅購買了 兩件商品的顧客同時買下，其意義遠大於被一位購買了上百件商品的顧客同時買下。For example：

- A customer who only buys mountaineering gear purchases both a hardshell jacket and a trekking pole (only 2 items) → The Adamic-Adar index assigns a high weight. 一位只買登山裝備的顧客同時買了衝鋒衣和登山杖 only 2 items → Adamic-Adar 給予高權重重；
- A "shopaholic" who buys across the entire site also purchases these two items → The weight is very low. 一位買遍全站的“剁手黨”也買了這兩件 → 權重很低。結果：系統更信任前者的行為，認為這是真實需求關聯。

Personalizing Affinity by User Segment 基於用戶細分的偏好個性化

Objective: Build separate **product affinity networks** for different customer segments (e.g., "high-value" vs. "budget") and compare recommendations. 為不同客戶群體（例如“高價值客戶”與“預算敏感型客戶”）分別構建**不同偏好的 產品親和網路**，並比較各群體的推薦結果。

Method

- Segment users (e.g., by total purchases) 首先根據使用者特徵（如總消費金額、購買頻次等）將用戶劃分為不同細分群體(例如“高價值” vs. “預算型”),
 - **Count** the number of transactions for each user in the dataset. **根據使用者活躍度進行分群（例如，按購買次數）**
 - Then, filter for users with 2 or more purchases, classifying them as "high-value" or "active" users.: 然後，篩選出購買次數 **大於等於 2 次** 的用戶，將其視為“高價值用戶”或“活躍用戶”
 - Reconstruct the bipartite graph (User-Product) specifically for this high-value user segment. **為這個高價值用戶群體重新構建二分圖（用戶-商品 bipartite graph）**

- For each segment, extract sub-bipartite graph 針對每個細分群體，中提取對應的**子二分圖** (sub-bipartite graph)
 - compute Jaccard/AA 在每個子網路上，使用相似性指標（如**Jaccard係數**或**Adamic-Adar指數**）衡量產品之間的關聯強度（即通過共同被該群體用戶購買/互動的程度反映偏好共現）。
- Recommend top products per segment. 基於 **相似性得分**，為每個細分群體推薦其最相關的**Top-K產品**。

該方法能夠捕捉不同使用者群體在 **產品組合** 偏好上的結構性差異，從而實現更精准、更具相關性的個性化推薦。

Substitutes vs. Complements via Network Signals 網路信號識別替代品與互補品

Use **purchase co-occurrence** (complements) vs. **browsing co-consideration** (substitutes). 利用使用者行為資料中的網路結構信號，區分產品之間的**互補關係** (complements)：傾向於被**同時購買**（如手機與耳機）與**替代關係** (substitutes)：傾向於被**共同流覽但很少一同購買**（如不同品牌的同類商品）

- **Complements**: 計算產品對之間的 **Jaccard 相似係數**（反映共同購買用戶的占比），High Jaccard on **purchase data**.
 - **高 Jaccard 值** → **強互補性**。但是 only 低 Jaccard 沒什麼功能
- **Substitutes**: 計算產品對之間的 **Adamic-Adar (AA)** 指數（更重視稀有共同流覽者，對長尾行為敏感）。
 - High Adamic-Adar on **view/browse** data (same users view both but rarely buy together). 若兩產品被相同使用者頻繁流覽，但極少被一同購買，則視為潛在替代品。**高 AA + 低 Jaccard** → **強替代性**

高 Jaccard 值 → **強互補性**，低 Jaccard 必須加上 AA，**高 AA + 低 Jaccard** → **強替代性**

「長尾」指**大量低頻出現、小眾但總量可觀的物品或用戶行為**，其分布符合**幂律**（Power Law）

This approach integrates outcome behaviors (purchases) with intent behaviors (browsing), effectively decoupling two distinct types of relationships: functional synergy (complementarity) and choice competition (substitutability). 此方法結合了**結果行為**（購買）與**意圖行為**（流覽），能有效解耦功能協同（互補）與選擇競爭（替代）兩類關係

It is highly applicable to product recommendation, category management, and the optimization of marketing strategies. 適用於推薦、品類管理和行銷策略優化。

Cold Start for New Products 新產品的冷開機問題

Estimate affinity for a new product **P5** using its **attributes** (e.g., category) and existing network. 利用新商品 **P5** 的**屬性資訊**（例如相同的商品品類）及其與現有商品的關聯，利用 **已有產品偏好網路** 估算其商品親和力。

Method

1. **屬性對齊**: Assign **P5** to a category (e.g., same as **P1**, **P2**). 將新商品 **P5** 歸入一個已知類別（例如與現有商品 **P1**、**P2** 同屬“無線藍牙耳機”類）
2. **親和力遷移**: Inherit affinities from similar products via **average Jaccard**. 基於同品類的已有商品（如 **P1**, **P2**），計算它們在使用者-產品二分圖中與 其他產品的 **Jaccard 相似度向量**（即每款產品對所有其他產品的 Jaccard 得分）。

- 然後，take the **average** of these similarity vectors to serve as the initial affinity representation for **P5** 對這些相似度向量取**平均值**，作為 **P5** 的初始親和力表示。
 - For example：If the Jaccard values between **P1** and products **A**, **B**, **C** are $[0.3, 0.1, 0.6]$, and the corresponding values for **P2** are $[0.4, 0.2, 0.5]$, then the initial affinity vector for **P5** would be $[0.35, 0.15, 0.55]$ 若 **P1** 與 **A**、**B**、**C** 的 Jaccard 值為 $[0.3, 0.1, 0.6]$ ，**P2** 對應值為 $[0.4, 0.2, 0.5]$ ，則 **P5** 的初始親和力向量為 $[0.35, 0.15, 0.55]$ 。
3. Based on this affinity vector, you can immediately identify potentially high-interest users for **P5** (e.g., users who have previously purchased or browsed products with high Jaccard similarity to **P5**), or include **P5** in recommendation lists for exploratory exposure 基於親和力向量，可立即為 **P5** 識別潛在高興趣用戶（如曾購買/流覽高 Jaccard 產品的使用者），或將其納入推薦列表進行探索性曝光。

Product Centrality → Business Impact 產品中心性對業務影響的分析

Identify **hub** products (**high degree in affinity network**) and test if they drive basket size. 識別親和網路中的 **樞紐型產品**（具有**高度中心性**的產品），並檢驗其是否能帶動**購物籃規模**（basket size）的提升

- Build product co-purchase graph **構建產品共購圖**: Connect every pair of products that appear in the same order to form a **single-mode product network** (a one-mode projection of the user-product bipartite graph) 將同一訂單中出現的 產品兩兩連接，形成一個**單模產品網路**（one-mode projection of the user-product bipartite graph）.
- then, compute **degree centrality** 對每個產品節點, 計算其**度中心性**（即與其他產品的連接數量）。A higher degree centrality indicates that a product is frequently purchased alongside other products, acting as a "hub" within the network 度中心性越高，說明該產品越常與其他產品一同被購買，是網路中的“樞紐”。

- Finally, correlate with average basket size when product is purchased **關聯業務指標：平均購物籃大小**:
 - For each product P_i , calculate the **average basket size** (i.e., the average number of items per order) across all orders containing P_i . 對每個產品 P_i ，計算所有包含 P_i 的訂單的**平均商品購買數量**（即 basket size）。

High-centrality products (e.g., P_1 , P_2) → larger baskets. **高中心性產品**（如 P_1 , P_2 ）通常作為“入口”或“基礎品”，常被搭配其他商品購買，因此其所在訂單的**平均購物籃更大**。

- If validated, these products can be leveraged for 若驗證成立，這類產品可被用於：
 - **Cross-selling Anchors**: Recommending high-centrality products to drive overall sales volume:推薦高中心性產品以拉動整體銷售額
 - **Promotional Strategy Prioritization**: Prioritizing the promotion of high-centrality items, as they are likely to generate a larger halo effect or spillover in associated sales:推廣 高中心性產品 可能帶來更大的連帶銷售效應

It is further beneficial to distinguish between weighted degree centrality (accounting for co-purchase frequency) and unweighted degree centrality to capture different types of hub effects. 可進一步區分**加權度中心性**（考慮共購頻率）與**未加權度中心性**，以捕捉不同類型的樞紐效應。

By controlling for factors such as category and price through regression analysis, one can more rigorously evaluate the marginal contribution of centrality to the overall basket size. 控制品類、價格等因素後做回歸分析，可更嚴謹地評估中心性對 basket size 的**邊際貢獻**。

Ethical Risks in Affinity Marketing 親和行銷中的倫理風險防控

Detect **risky product pairs** (e.g., diet pills 減肥藥+ laxatives 瀉藥，通便劑) that should **not** be recommended together. 具有潛在倫理或健康風險的產品組合（例如“減肥藥 + 瀉藥”），避免將此類高親和度產品對推薦給使用者，防止造成身體傷害、誤導消費或引發合規問題。

1. **Flag** sensitive categories (manual or keyword-based) 敏感品類標記：通過**人工規則或關鍵字匹配**（如“減肥”、“瀉藥”、“精神類”、“醫療器械”等）建立敏感產品類別清單。
2. **高親和商品對篩查**: If both products in a high-affinity pair are sensitive → suppress recommendation. 在高度親和產品網路中，識別出 Jaccard 或 Adamic-Adar 相似度較高的 產品組合。檢查這些高親和度產品組合對是否**雙方均屬於敏感類別**。
3. **推薦壓制策略**（Suppression Rule）: If **both products** in a high-affinity pair are **flagged as sensitive**, the recommendation for this combination must be **blocked or suppressed**, even if the data indicates a strong association. 若一個高親和度產品對中的**兩個產品均被標記為敏感**，則**禁止或壓制**該組合的推薦（即使資料上顯示強關聯）。

- Risk Combination Example: Diet Pills + Laxatives：減肥藥 + 瀉藥
 - Although users may frequently purchase these together (resulting in a high Jaccard index), their combined use poses significant health risks 雖然用戶可能頻繁同時購買（高 Jaccard），但聯合使用存在健康隱患。
 - Even if the model suggests "users who like A also like B," this combination should be proactively **not recommended** 即使模型認為“喜歡A的人也喜歡B”，也應主動**不推薦**此組合。

Application in Social Media Analysis

Bipartite Network: *Users* ↔ *Hashtags / Groups / Events* 使用者 ↔ 話題/標籤/群組

- **User用戶** represent one type of node (people個人, accounts帳號, influencers意見領袖) 是一類節點；**Affiliations** (hashtags話題標籤, topics主題, groups群組, events事件, or even URLs/domainsURL/功能變數名稱) represent another；
- A connection is formed whenever a user **posts, reposts, or engages** with a topic. 當某使用者 **發佈、轉發或參與某個** 話題時，就形成一條連接。例如：
 - ■ User A publishes a post with #CarbonNeutrality → Creates a link: A — #CarbonNeutrality 用戶A 發了帶 #碳中和 的帖子 → A — #碳中和
 - User B uses both #CarbonNeutrality and #GreenFinance simultaneously → This implies a semantic or behavioral association between these two tags. 用戶B 同時使用了 #碳中和 和 #綠色金融 → 暗示這兩個標籤存在語義或行動上的關聯。

Marketing Concept	Social Media Equivalent
Product	Hashtag, Topic, Group
Customer	Social Media User
Co-purchase	Co-mention / Co-participation
Product Affinity	Hashtag Affinity or Ideological Proximity

Metrics in Action:

- **Jaccard Similarity** between hashtags: 標籤親和力 (Hashtag Affinity) : Measure the strength of their association in public perception by calculating the frequency with which two tags are co-used by the same user 通過計算 兩個標籤組被同一使用者共同使用的頻率，衡量它們在公眾認知中的關聯強度。
How often do #ClimateAction and #GreenNewDeal appear in the *same posts* relative to their total usage? High Jaccard → tightly linked narratives. Jaccard相似度：排除熱門標籤的“噪音”，聚焦真正共現比例高的組合. 例如：#環保 與 #素食 在小眾群體中高頻共現，Jaccard值高，說明存在特定價值觀社群。
- **Adamic-Adar** for user influence: If two hashtags are co-used by users who *rarely post*, that signal is stronger (e.g., activists vs. bots). **Adamic-Adar指數**：更重視“低活躍但專注”的用戶（如環保活動家）的共現行為，而非大V的泛化發帖。

- **Common Neighbors 共同鄰居:**

Number of users who engage with both a brand and a social movement → reveals brand alignment risks/opportunities. 同時參與某個品牌和某項社會運動的用戶數量 → 揭示該品牌在價值觀契合方面的風險或機會

Identify **bridge users** who connect polarized communities (e.g., pro/con vaccine groups) — critical for moderation or outreach. 識別連接立場對立社群的 **橋樑用戶**（例如，支持與反對疫苗接種的群體）——這對內容審核或外聯策略至關重要。

1. **Community Detection** and **Bridge Identification** 社區發現與橋接者識別: By **projecting** the bipartite network into a Tag-Tag Network or a User-User Network. 將二分網路 **投影** 為 標籤-標籤網路 或 使用者-使用者網路，可識別：
 - Discourse Clusters **話語集群** ("Climate Activists" vs. "Tech Optimists “氣候行動派” vs “技術樂觀派”)；Cross-Boundary Influencers **跨界影響者**：Users who engage in multiple seemingly unrelated topics (e.g., discussing both #ESG and #SupplyChain), acting as critical information bridges同時參與多個看似無關話題的使用者（如既談 #ESG 又談 #供應鏈），可能是關鍵資訊橋樑。
2. **Risk and Opportunity Insights** 風險與機會洞察
 - Reputation Risk: If a brand unexpectedly co-occurs with sensitive tags (e.g., #BoycottXX), it may signal an impending reputation crisis 品牌若與某些敏感標籤（如 #抵制XX）出現意外共現，可能預示聲譽風險；
 - Emerging Narratives: New tag combinations (e.g., #RegenerativeAgriculture + #CarbonCredits) may represent the next wave of sustainable investment narratives. 新興標籤組合（如 #再生農業 + #碳信用）可能代表下一波可持續投資敘事。

應用場景：**Monitoring** the evolution of public sentiment on environmental issues, identifying misinformation propagation paths, and evaluating the effectiveness of corporate ESG communication.

Detecting Narrative Clusters via Hashtag Affinity 通過話題標籤親和性識別敘事簇

Identify tightly linked hashtag groups that represent **coherent narratives** (e.g., “climate justice” vs. “energy security”). 從社交媒體資料中識別出 **緊密關聯的話題標籤 (hashtag) 群組**，這些群組共同構成具有語義一致性的**公共敘事**（如“氣候正義” vs. “能源安全”），從而揭示不同話語體系或輿論陣營。

- First, construct a User-Hashtag Bipartite Network 構建**使用者-話題標籤**二分網路: Nodes include Users (Mode A) and Hashtags (Mode B). An edge represents a user employing a specific hashtag in a post 節點包括：使用者（Mode A）與話題標籤（Mode B）。邊表示某用戶在某條帖子中使用了某個話題標籤。

- Project to a Hashtag-Hashtag Network (Hashtag Affinity Network) 投影為 話題標籤-話題標籤網路 (Hashtag Affinity Network)
- Project to **hashtag-hashtag** network using **Jaccard similarity** to reduce noise from viral hashtags. 將二分圖單模投影到話題標籤層面，衡量任意兩個標籤的共現強度。採用 **Jaccard 相似係數**（而非簡單共現頻次）以抑制病毒式熱門標籤的雜訊干擾。
 - Jaccard similarity is more sensitive to rare but highly overlapping tag combinations, helping to highlight consensus narratives within communities rather than generalized public trends. Jaccard 對稀有但高度重疊的標籤組合更敏感，有助於突出**社群內部共識性敘事**，而非大眾泛化熱點

- Clustering 識別敘事簇: Cluster using modularity or thresholding 。
 - **Modularity Maximization 模組度優化**: Utilize algorithms such as **Louvain** or **Leiden** to automatically discover highly cohesive topic communities by retaining strong connections and pruning weak ones 使用 Louvain 或 Leiden 等演算法，在保留強連接、切斷弱連接的前提下，自動發現高內聚的話題社區
 - **Similarity Thresholding 相似度閾值切割**: Retain only edges with a **Jaccard value** above a specific threshold (e.g., 0.1, or determined via statistical significance testing), then extract connected components as narrative clusters. 僅保留 Jaccard 值高於某閾值（如 0.1 或經統計顯著性檢驗確定）的邊，然後提取連通子圖作為敘事簇
- **Narrative Interpretation and Naming 敘事解釋與命名**: Perform thematic summarization on the hashtags within each cluster (using methods like keyword co-occurrence, word embedding clustering, or manual annotation) to assign interpretable narrative labels (e.g., "Climate Justice," "Green Tech," "Energy Sovereignty"). 對每個簇內的標籤進行主題歸納（如關鍵字共現、詞嵌入聚類、人工標注），賦予其可解釋的敘事標籤（如“氣候正義”、“綠色技術”、“能源自主”等）。

Tracking Emerging Hashtags via Adamic-Adar 通過 Adamic-Adar 指數追蹤新興話題標籤

Spot rising hashtags that are co-used by **focused/niche users** (not mass broadcasters), signaling authentic grassroots momentum. 識別正在興起的、由**專注型或小眾用戶**（而非大眾廣播者或網紅）共同使用的話題標籤（hashtags），以此捕捉真實、自下而上的**草根動量**（grassroots momentum）。

Why Adamic-Adar? It gives more weight to co-occurrence by users who post **few hashtags** (likely intentional), vs. influencers who spam many. 在使用者-話題標籤二分網路中，兩個標籤的共現可能源於兩類截然不同的行為。

- **Focused Users 專注用戶**：These users post less frequently and employ hashtags judiciously. Consequently, the tags they co-use often reflect clear intent and semantic consistency. 發佈內容較少、使用標籤謹慎，其共用標籤往往具有**明確意圖和語義一致性。
- **High-Volume/Broadcaster Users 高產/廣播型用戶**（如網紅、行銷帳號）：These users deploy a large volume of hashtags to maximize exposure. This behavior often results in incidental or strategic co-occurrences that lack deep semantic association. 大量使用標籤以擴大曝光，導致偶然性或策略性共現，缺乏深層關聯。

The Adamic-Adar Index favors the former group. **Adamic-Adar 指數** 偏好前者。 This is because it assigns **higher weight** to co-occurrences between users who publish fewer hashtags—such users are more likely to employ tags with clear intent. 因為它對**發佈較少話題標籤的使用者**之間的共現賦予更高權重——這類用戶使用標籤更可能是出於明確意圖；

In contrast, high-volume or influential accounts often use a large number of hashtags ("hashtag stuffing"), resulting in co-occurrence signals with greater noise. By penalizing the contribution of high-frequency neighbors (i.e., active hashtagging users), Adamic-Adar effectively highlights high-quality co-occurrence signals from focused users. 相比之下，高產或影響力大的帳號往往大量使用標籤（“刷標籤”），其共現信號雜訊更大。Adamic-Adar 通過懲罰高頻鄰居（即活躍發標用戶）的貢獻，有效突出了來自 **專注使用者** 的高品質共現信號。

A high AA value reflects hashtag co-occurrence driven by low-frequency, high-intent users, making it more likely to represent genuine community consensus or emerging narratives. **高 AA 值** 反映的是由**低頻、高意圖用戶**驅動的標籤共現，更可能代表真實的社群共識或新興敘事

Signal

- High AA → meaningful alignment; low AA → accidental or spammy co-use ; High Adamic-Adar (AA) Value: Indicates that hashtag co-occurrence is driven by intentional niche users. This signals authentic grassroots momentum and warrants attention. 高 Adamic-Adar (AA) 值 → 標籤共現由**有意圖的小眾用戶**驅動 → **真實草根動量**，值得關注
- Low AA Value: Suggests co-occurrence stems primarily from high-volume broadcasters or random combinations. This is likely noise, marketing manipulation, or a fleeting trend, often resulting from accidental overlap or spammy/flooding usage (e.g., random or strategic hashtag stuffing by mass-posting accounts) 低 AA 值 → 共現主要來自**高產廣播者**或隨機組合 → 可能是噪音、行銷或短暫熱點. 更可能為**偶然共現或垃圾式/氾濫式使用**（如由大量發標用戶隨機或策略性堆砌標籤所致）。

Disinformation Detection: Coordinated Inauthentic Behavior 虛假資訊檢測：識別協同性非真實行為

Find groups of accounts that **unnaturally share the same niche hashtags**—a sign of bot coordination. 發現一組帳號在**小眾/低頻話題標籤**上表現出異常一致的使用模式——這是自動化水軍（機器人啦 bots）或人工操控帳號進行 協同傳播（coordinated campaigning）的關鍵信號。

Signal 核心信號與指標

- High **Common Neighbors** on **rare hashtags** 在 **罕見話題標籤** 上具有高度共同鄰居（High Common Neighbors on Rare Hashtags）。In a user-hashtag bipartite network, if multiple accounts repeatedly co-appear under the same set of low-frequency hashtags (e.g., **#ObscurePolicy2026**, **#NicheTechX**)—tags that are rarely used by the general population—their behavior **deviates from the randomness expected in natural grassroots participation**. 在使用者-標籤二分網路中，若多個帳號**反復共同出現在同一組低頻標籤下**（例如 **#ObscurePolicy2026**、**#NicheTechX**），而這些標籤在整體用戶中極少被使用，則表明其行為**不符合自然草根參與的隨機性**。
- A high number of **Common Neighbors (CN)** indicates that these accounts form a "**dense subgraph**" connected via these tags, suggesting pre-scripted activity or centralized coordination. 共同鄰居數量高 → 帳號之間通過標籤形成“密集子圖”，暗示**預設腳本或集中指揮**

- Low diversity in hashtag usage per user **用戶標籤使用多樣性極低**: Real users typically employ a **diverse range of hashtags** across different posts, reflecting broad interests or contextual variations 真實用戶通常在不同帖子中使用**多樣化標籤**，反映興趣廣度或上下文變化。
- Inauthentic accounts tend to **repetitively use fixed hashtag combinations** (e.g., always posting `#TruthNow + #WakeUpPeople + #HiddenAgenda`), resulting in significantly lower **hashtag entropy** or a low ratio of unique tags 水軍帳號則傾向於**重複使用固定標籤組合**（如始終發佈 `#TruthNow + #WakeUpPeople + #HiddenAgenda`），導致其標籤熵（entropy）或唯一標籤占比顯著偏低. Individual accounts exhibit extremely low topic diversity, repeating a narrow, fixed set of tags without the contextual variation or interest breadth seen in real users 單個帳號的話題標籤使用多樣性極低：每個用戶僅重複使用固定、狹窄的一組標籤，缺乏真實用戶常見的語境變化或興趣廣度。

為何這兩項指標結合有效？結合這兩項指標，可有效甄別偽裝成草根參與、實則由自動化或人工水軍操控的協同傳播活動。

指標	單獨使用風險	聯合優勢
高 Common Neighbors	可能是真實小眾社群（如專業學者討論冷門議題）	若同時標籤多樣性低 → 排除真實興趣群體
低標籤多樣性	可能是專注型真實用戶（如只討論環保）	若同時與多人共用罕見標籤 → 揭示非自然協調

✔ **Joint Criterion:** A group of accounts using nearly identical, few tags, repeatedly appearing under the same obscure topics" → ****Highly likely to be an inauthentic influence operation **** (astroturfing/botnet). “一群帳號用幾乎相同的少數標籤，反復出現在同一冷門話題下” → 極可能為水軍。

- **Filter for Low-Frequency/Niche Hashtags**篩選低頻/小眾標籤:
 - Define "**Rare Hashtags**": Tags used by fewer than a specific threshold of users globally (e.g., the bottom 5% by frequency).全網使用用戶數 < 閾值（如前5%最低頻標籤）。
 - These tags possess **high semantic specificity**; the probability of organic overlap among normal users is naturally low這些標籤天然具有**高語義特異性**，正常用戶重合概率低。
- **Construct a Suspicious Sub-network**構建可疑子網路:
 - Extract all users who have engaged with these rare hashtags and their associated tag sets提取所有使用過罕見標籤的使用者及其標籤集合。
 - Calculate the ****Common Neighbors**** (CN) count between user pairs (i.e., the number of rare hashtags shared by two users 計算用戶對之間的 **Common Neighbors** (CN) 數量（即共同使用的罕見標籤數）。

- **Identify High-Synergy Groups** 識別高協同群體: For user pairs with high CN values, conduct further checks on 對 CN 值高的用戶對，進一步檢查其：

§ **Hashtag Diversity**: (e.g., Unique Tags / Total Posts) 標籤多樣性（如唯一標籤數 / 總發帖數） / § **Temporal Synchronicity**: Whether posts were concentrated within a short time window 時間同步性（是否在短時間內集中發帖） / § **Content Similarity**: Text repetition rates or use of templated language 內容相似度（文本重複率、範本化語言）

- **Clustering and Labeling** 聚類與標記:

- Apply community detection algorithms (e.g., **Louvain**) to identify tight clusters within the high-CN user subgraph. 使用社區發現演算法（如 Louvain）在高 CN 用戶子圖中識別緊密集群。
- **Flagging Criteria**: If a cluster exhibits Low Diversity + High Common Neighbors + High Temporal Synchronicity, label it as "Suspected Coordinated Inauthentic Behavior." 若集群內用戶普遍呈現低多樣性 + 高 CN + 時間同步，則標記為“疑似協同非真實行為”。

Application in Finance

Translate the **Product Affinity Network** to financial markets to **monitor systemic risk, identify herding behavior, and reveal latent correlations beyond price movements**. 將產品親和網路 (Product Affinity Network) 遷移至金融市場，用於**監測系統性風險、識別羊群行為 (herding)**，並揭示價格相關性之外的隱性關聯。

Bipartite Network: *Investors*投資者 ↔ Stocks or Funds ↔ Assets 資產 (股票/債券/基金)

- **投資者** (如對沖基金、養老基金、個人大戶) 是一類節點；**金融資產** (如特斯拉股票、某綠色債券、比特幣) 是另一類節點；
 - **Mode A (Actors):** Institutional investors (hedge funds, mutual funds, pension funds), or even retail investor clusters
 - **Mode B (Affiliations):** Stocks, bonds, ETFs, cryptocurrencies, or companies
 - **Edge:** Exists if an investor holds the asset (from 13F filings, portfolio disclosures, or brokerage data)

- When an investor holds an asset, a connection is established 當某投資者持有某資產，即建立連接。
例如：
 - **Bridgewater Associates** holds both **Apple** and **Microsoft** → Creates edges:
Bridgewater – Apple and Bridgewater – Microsoft § 橋水基金（是全球最大的對沖基金） 持有 蘋果和 微軟 → 橋水 – 蘋果，橋水 – 微軟
 - If multiple funds simultaneously hold significant positions in **TSMC** and **ASML** → This implies a '**consensus allocation**' within the semiconductor equipment/value chain sector. 若多家基金同時重倉 台積電 和 阿斯麥 → 暗示半導體設備/價值鏈板塊存在"共識性配置"。

Goal: Monitor systemic risk, detect herding, or uncover hidden correlations.

Marketing Concept	Finance Equivalent
Product	Stock, Asset, Company
Customer	Investor, Hedge Fund, Mutual Fund
Co-purchase	Co-holding (same investor owns both assets)
Product Affinity	Portfolio-based asset correlation

Core Idea: The "**affinity**" between assets stems not only from price co-movements but also from being **strategically held by the same investor groups**. 資產之間的“親和力”不僅來自價格聯動，更源于被相同投資者群體戰略性持有。

Metrics in Action:

- **Common Neighbors:** How many funds hold both Stock A and Stock B? High count may indicate **herding behavior**. 同時持有股票 A 和 B 的基金數量。信號：若該數值異常高，可能反映**羊群行為**（herding）——大量機構同步押注相似資產組合，放大市場脆弱性。風險提示：在流動性緊張時，任一資產遭拋售可能觸發連鎖反應
- **Jaccard Similarity:** Normalized overlap in investor bases → reveals **fundamental linkage** beyond price correlation (e.g., ESG-focused funds holding both renewable energy firms). 歸一化處理，避免大市值資產因天然被廣泛持有而產生虛假關聯。洞察：高 Jaccard 值揭示**基本面或策略層面的深層聯繫**（如 ESG 基金群同時重倉兩家可再生能源公司），即使二者股價短期不相關。

- **Adamic-Adar:** If two obscure stocks are held together by a few specialized hedge funds (not mega-funds), that's a stronger signal of **strategic linkage** than if they're held by BlackRock (which owns everything). 若兩家冷門股票被**少數專注型對沖基金**（持倉集中）共同持有 → AA 值高 → **強戰略關聯信號**; 若僅被 BlackRock 等“全市場覆蓋型”巨無霸基金持有）→ 貢獻被大幅衰減 → 不視為有效關聯. **價值：**過濾“被動共現”，聚焦**主動、有資訊含量的協同持倉**

Use case: During market stress, high co-holding similarity can predict **contagion risk**—if one stock drops, forced selling may spill over to its "affiliated" peers. 若 A 與 B 具有**高共持相似性**（高 Jaccard 或 AA），則 B 很可能被**連帶拋售**，即使其基本面未受影響。**預警：**提前識別高親和資產對，構建“**持倉網路傳染圖譜**”，輔助壓力測試與風險隔離。

Application of Similarity Metrics :

- **Common Neighbors (共同持有人數量)** : Quickly identifies "**crowded trades**" (stocks held by many of the same funds)快速識別“抱團股”，Highly susceptible to noise from **large index funds** or ETFs that hold everything, creating false signals of affinity 但易受大型指數基金干擾。
- **Jaccard相似度** : Measures the **proportion of overlap** in the investor base, effectively filtering out **scale bias**衡量兩資產 投資者基礎的重合比例，過濾規模偏差。
 - “ Stock A and Stock B are both held by **5 funds**. If Stock A is held by 6 funds total and Stock B by 7股A和B被5家基金同時持有，總持有基金數分別為6和7 → $Jaccard = 5/(6+7-5) = 0.625$ ，關聯性強。The **high ratio** indicates a very strong strategic link between these assets ”
- **Adamic-Adar指數** : Places higher weight on co-holdings by **specialized/niche investors** (e.g., a fund that *only* invests in solar energy)更看重**專業型投資者**（如only 只投太陽能的基金）的共同持倉，認為其信號更可信。A co-holding signal from a **focused expert** is considered more significant and credible than one from a diversified generalist. It downweights the "noise" of broad-market holders and highlights **high-conviction thematic links**。

Asset Affinity Network for Alternative Risk Factors 基於共同持倉的資產親和網路：挖掘另類風險因數

Discover **non-price-based relationships** between assets (e.g., ESG alignment, thematic investing) using co-holding patterns. 通過機構投資者的**共同持倉行為**（co-holding patterns），構建**資產親和網路**，以發現超越價格相關性的**非價格驅動關聯**——如 ESG 一致性、主題投資邏輯（如“人工智慧”、“清潔能源轉型”）或戰略敘事對齊。

Why? Two stocks may have low price correlation but be held by the same ESG funds → they share **fundamental or strategic linkage**. 兩支股票即使價格走勢無關（甚至負相關），若被同一類專注型基金（如 ESG 基金、氣候主題基金）系統性地同時持有，則它們在**基本面邏輯、風險暴露或投資敘事層面**存在深層聯繫。

Use **Jaccard Similarity** to normalize for popular assets (e.g., AAPL is held by everyone). 採用 **Jaccard 相似度** 計算資產對之間共同持倉強度. 該指標能有效**歸一化普遍持倉偏差**（例如 AAPL 幾乎被所有基金持有），避免高流通性或熱門資產主導相似性結果，從而凸顯真正具有共同投資邏輯的資產對。

- **The Blind Spot:** Price correlation fails to capture asset linkages driven by **investment philosophy, regulatory constraints, or strategic preferences**. 價格相關性無法捕捉由**投資理念、監管約束或策略偏好**驅動的資產聯動。Two assets may move independently in normal markets but crash together when the specific investor base holding them reacts to a common trigger
 - 股票 A（風電設備商）與股票 B（綠氫技術公司）價格相關性低；但均被“淨零碳排”主題基金重倉 → 在政策利好/利空 ESG 時可能**同步遭遇資金流入/流出**。風險含義：此類隱性關聯構成**另類系統性風險因數**（如“ESG 擠兌風險”），在市場壓力下可能引發跨資產傳染。

關鍵優勢：Automatic Normalization of "Popular Asset" Bias自動歸一化熱門資產偏差: **Apple (AAPL)** is held by almost every fund. Consequently, its **raw co-holding count** with *any* other asset is artificially high, creating **false associations** (noise). 蘋果（AAPL）被幾乎所有基金持有 → 其與任何資產的**簡單共持數量**都很高，造成虛假關聯。By normalizing against the **union** (total unique holders), Jaccard penalizes pairs involving ubiquitous assets unless the overlap is proportionally massive. Jaccard 通過**並集歸一化**，有效抑制此類偏差。

Systemic Risk via Co-Holding Contagion 共持傳染識別系統性風險

Identify asset pairs at risk of **fire-sale contagion**: if one drops, forced selling spills to its high-affinity partners. 識別因**機構共同持倉**而存在**拋售傳染**（fire-sale contagion）風險的資產對——即當某一資產價格暴跌引發強制平倉時，其高親和度的關聯資產可能因被同一群投資者持有而遭遇**連帶拋售**，即使二者基本面或價格走勢原本無關。

共持即耦合，耦合即傳染通道

- Under market stress conditions (e.g., **liquidity crunches**, **massive redemptions**, or **margin calls**), institutional investors are often forced to raise cash rapidly 在市場壓力下（如流動性緊縮、大規模贖回、保證金追繳），機構投資者往往**無差別減持其持倉組合中的資產**以快速回籠現金。
- Instead of selectively selling based on fundamentals, they engage in **indiscriminate deleveraging**, reducing positions across their entire portfolio to meet liquidity needs. 若資產 A 與 B 被大量相同機構持有（**高共持數量**），則 A 的下跌可能觸發對 B 的**非理性或被動拋售**，形成“**流動性驅動的跨資產溢出**”。
- Assets are sold not because they are "bad," but simply because they are **held**. 這種風險**獨立於價格相關性**：即使 A 與 B 歷史回報不相關，只要它們在投資組合中“綁在一起”，就構成隱性系統性脆弱點。

Use **Common Neighbors** (raw co-holder count) as a proxy for liquidity linkage. High count = higher spillover risk. 以 Common Neighbors 作為流動性耦合代理指標。

- **Common Neighbors (CN) as a Risk Proxy : High CN Value:** Indicates tight **portfolio coupling**. The assets are "bundled" together in the balance sheets of many major players。CN值越高 → 投資組合層面耦合越緊密 → 流動性衝擊下的拋售傳染風險越高。
- **Risk Implication:** Higher CN directly correlates with a higher probability of **sell-off contagion** during liquidity shocks. If one asset fails, the shared holders are the transmission mechanism for the crash。 **風險排序與預警:** 對全市場資產對按 CN 值排序，識別“高共持但低價格相關”的異常組合。在市場波動率上升或特定資產出現異常資金流出時，
優先監控其高-CN關聯資產。

When market volatility spikes or a specific asset experiences abnormal capital outflows, **prioritize monitoring its high-CN partners**。一旦其中一資產遭遇流動性衝擊（如大幅下跌觸發贖回或保證金追繳），持有該資產的機構可能被迫同步減持其關聯資產，從而**放大跨資產的拋售溢出風險**。此方法尤其適用於監測系統性風險在非價格相關但持倉高度重疊的資產間的隱性傳導路徑。

Uncovering "Shadow Correlation" via Ownership Structure

- **Asset A:** Regional Bank Stock：某區域性銀行股（Currently declining due to specific real estate exposure risks**近期因地產風險下跌）；****Asset B:** An Infrastructure REIT（Fundamentals remain robust; traditionally **uncorrelated** with the banking sector in terms of industry classification or price history. 基本面穩健，與A行業無關）
 - $CN(A,B)=120$, 120 家共同持有機構
 - Asset A continues to fall, triggering a wave of **fund redemptions** and **margin calls** for the shared holders, these 120 institutions engage in **synchronous selling**. They are forced to liquidate Asset B 這120家機構可能同步減持B以補流動性 → not because its fundamentals have changed, but simply because it is a liquid asset in their portfolio that can be sold to cover losses in Asset A. B價格非理性下跌 → 形成跨行業傳染。

Shadow Correlation 隱藏在所有權結構中的“影子相關性”，It transforms risk management from reacting to price moves to anticipating structural vulnerabilities 為防範“黑天鵝”式連鎖反應提供早期預警能力。

Detecting "Smart Money" Signals with Adamic-Adar 用 Adamic-Adar 指數識別“聰明錢”信號

Find asset pairs linked by **specialized investors** (not mega-funds), which may signal alpha. 通過機構持倉網路，發現由**專業化、高集中度投資者**（如主題型對沖基金、主動管理人）共同持有的資產對，這些關聯更可能蘊含**前瞻性觀點或超額收益信號**（alpha），而非被動配置噪音。

Why Adamic-Adar?

- A high **Adamic-Adar score** between two assets indicates that they are linked by **professional, active investors** who have made a conscious decision to pair them 基金持倉越集中（ $|N(f)|$ 越小），其作為“連接橋樑”的貢獻越大。

Adamic-Adar transforms co-holding data from a simple count of "who owns what" into a **quality-adjusted map of investor conviction**, isolating the strategic moves of experts from the passive drift of the crowd Adamic-Adar 本質是資訊稀有性加權：越少見的共持行為，權重越高。因此，高 AA 值的資產對更可能反映了**由專業、主動型投資者驅動的戰略性關聯**，是挖掘潛在 alpha 的有效線索。